

Volume profile, TPO, EMA and ATR features on the 07:00–11:00 Donchian scalp

Forty-five causal features tested as entry filters, take-profit levels and stop variables on a coin-flip breakout base. One condition survived every gate: a prior-session TPO single print within 3 ATR above the close. Everything the brief named by hand — EMA200 as support, the EMA 13/48 cross, profile levels as targets, ATR-scaled stops — failed its matched control. This document is written for a second quant to audit and to decide what to do next.

1.110 →
1.382

locked profit factor,
base → gated
(+24.5%)

p 0.016

locked, vs a random
filter of the same
selectivity

187

locked trades (389
before the gate)

0.056

locked bootstrap
P(mean ≤ 0) — not
yet clear of zero

1 / 43

entry conditions
surviving BH q 0.10

2.8×

MC p99 drawdown
over realised

1. What the specialist needs to know first

- **Data.** NQ continuous futures, 1-minute bars 2022-12-26 to 2025-12-12 (70,685 15-minute bars), New York clock, extended hours. Price *levels* are synthetic (ratio to the real index decays 1.25 → 1.02 over the sample), so returns, R and ATR units are trustworthy and dollar figures are inflated ~12% early. Bid/ask is not in the feed; spread is assumed.
- **Blocks.** Research = first 65% of sessions, to 2024-11-27. Locked = 2024-11-27 to 2025-12-12, read exactly once, after every choice was fixed. Nothing after 2025-12-12 exists in the file.

- **Costs.** MNQ: \$0.72 commission per side plus one tick of slippage per side, charged inside the walker on every fill. One contract, no compounding, results in percent of entry price.
- **Base strategy (frozen before this study).** Long only. Donchian 10-bar entry (high above the prior 10-bar high, fill at the next open), 10-bar channel exit, $3.19 \times \text{ATR}(14, \text{EMA of true range})$ stop anchored at the fill, $2.3 \times \text{ATR}$ target, hard cap 230 minutes, entries 07:00–11:00 New York, one position at a time. It is the surviving finalist of a 3,600-trial Optuna search under those constraints (docs/ib/STUDY_BAYESOPT_SCALP.md) and had already been shown to clear a random entry in the same window on both blocks (locked p 0.040). A 38-condition indicator pool had already failed to raise its locked PF (docs/ib/STUDY_SCALP_FILTERS.md): best in-sample lift +12%, the only BH survivor ($\text{ADX} \geq 20$) inverted out of sample.
- **Discipline.** Every feature passed a truncation audit (recompute on history ending at bar i , value at i must match) and the volume-profile module's own leakage check. Base rates of every condition were measured on the trigger's own bars before any P&L. Every entry condition was scored against 250 random filters keeping the same number of signal bars; every level target against 200 random targets drawn from its own distance distribution; the stop axis reported in %/trade, R and total-at-one-unit with drawdown because those three disagree on this branch. 152 things were looked at on research before the locked block was opened.

2. Headline result

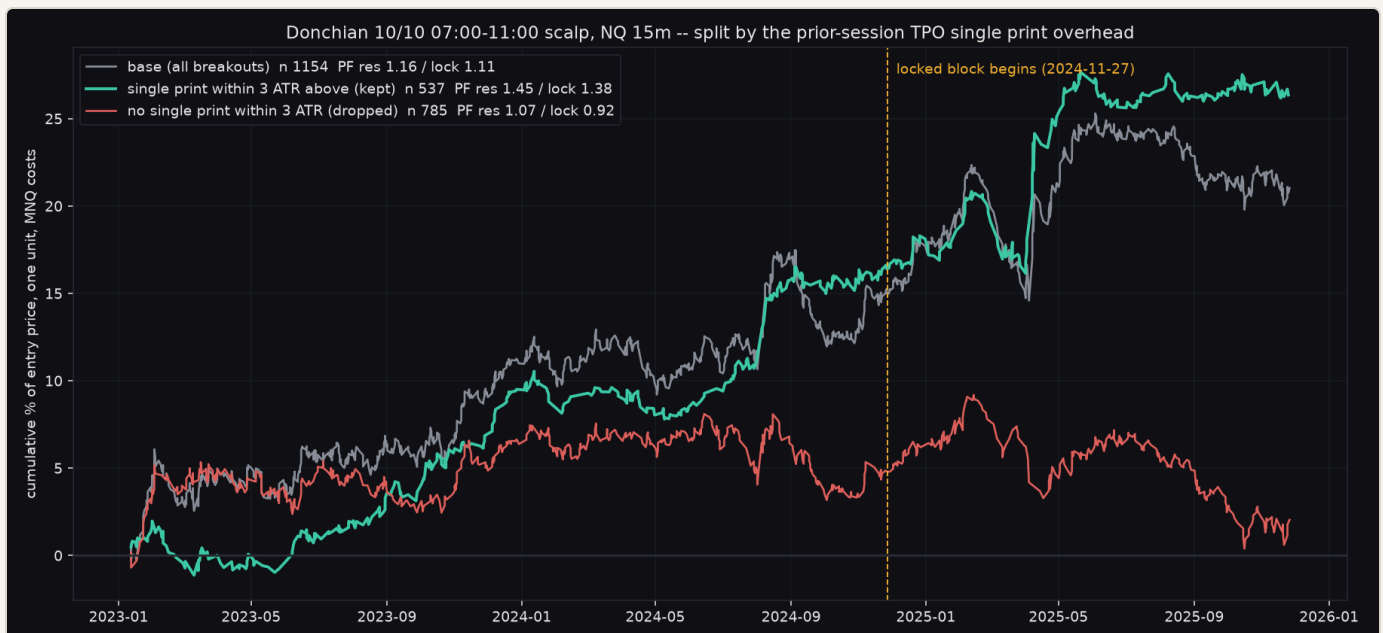


Fig 1. Cumulative return in percent of entry price. Grey: the base. Green: trades with a prior-session single print within 3 ATR above the close. Red: the rest. Kept and dropped are each run with their own position lock, so their counts sum to more than the base.

	N	PF	WIN	TOTAL	SHARPE	MAX DD	R/TRADE	NULL
base, research	765	1.164	51.5%	+15.0%	1.26	5.5%	+0.026	
gated, research	350	1.446	52.0%	+16.4%	2.84	—	+0.085	random filter median 1
base, locked	389	1.110	48.8%	+6.0%	0.73	7.8%	-0.008	
gated, locked	187	1.382	50.8%	+9.9%	2.02	4.7%	+0.053 %/tr	random filter median 1

It decays across the split (1.446 → 1.382), which is the shape an edge should have; it is positive in every calendar year (2023 PF 1.52, 2024 1.43, 2025 1.33 against the base's 1.24 / 1.14 / 1.07); it earns more total money on half the trades. Sharpe here is annualised from daily sums over every trading day, zero-filled on days that did not trade.

3. The mechanism



Fig 2. A locked-block trade. Left: the prior session's 15-minute bars and its TPO profile (30-minute letters, 2.5-point bins); orange bins were touched by exactly one letter. Right: the next morning's

breakout at 10:30 with the nearest single print 2.07 ATR overhead and the 2.3 ATR target just above it.

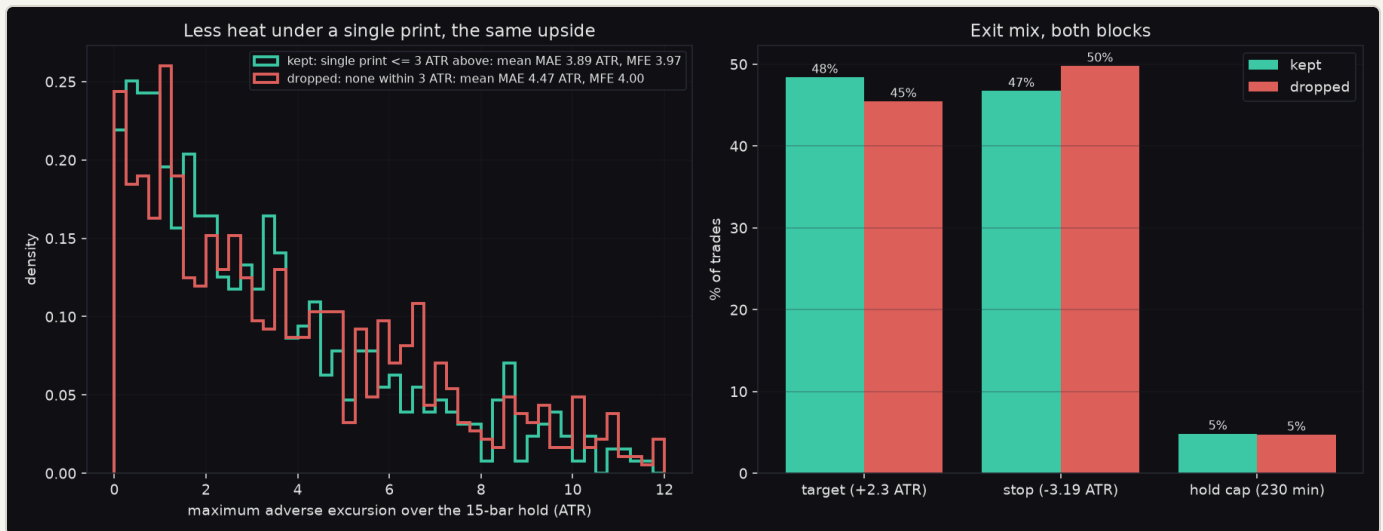


Fig 3. Fixed-horizon adverse excursion over the 15-bar hold, kept vs dropped, and the exit mix on both blocks.

Inside the condition the mean MAE is **3.69 ATR against 4.44** outside, with MFE unchanged (4.11 / 4.06). The target is reached 50% of the time against 47%, the stop 45% against 49%. The reading is auction-theory: a single print marks a price zone the prior session traversed inside one letter, i.e. no acceptance, and a breakout heading into it meets less overhead supply. The only strong correlate of the single-print distance is the distance to the prior session's high ($p = -0.49$), which is geometry, and the drop-one in \$5 shows that geometry is not the carrier.

4. The distance ladder

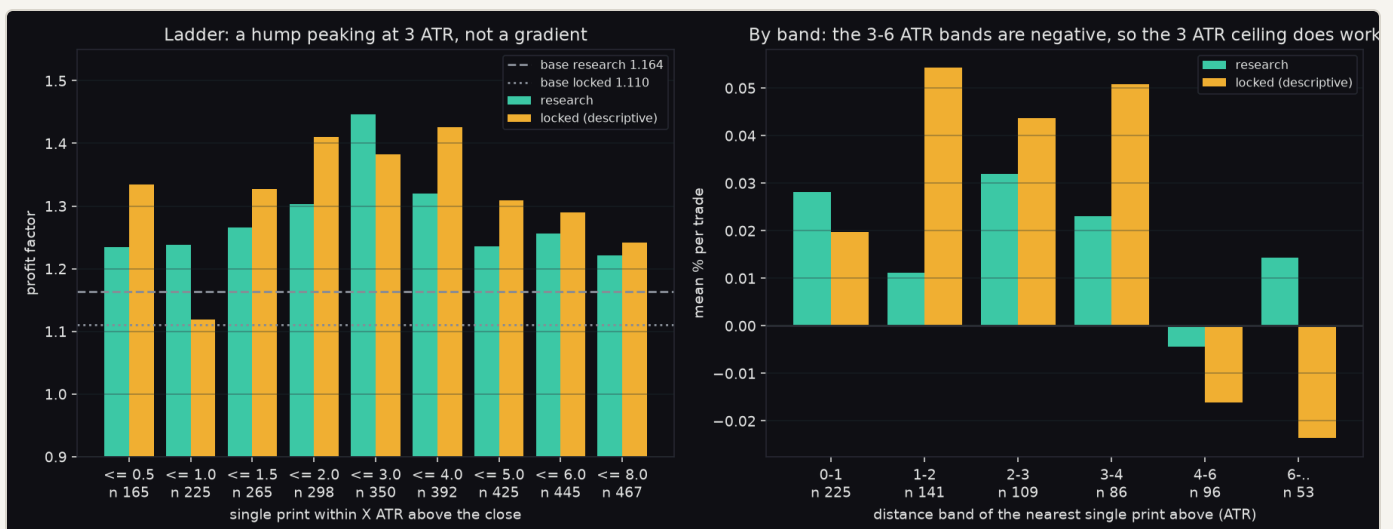


Fig 4. Left: PF by ceiling on the single-print distance, research and locked. Right: mean %/trade by distance band.

WITHIN X ATR	0.5	1.0	1.5	2.0	3.0	4.0	5.0	6.0	8.0
research PF	1.234	1.238	1.266	1.303	1.446	1.320	1.235	1.256	1.222
research control p	0.175	0.120	0.075	0.010	0.000	0.000	0.060	0.040	0.095
locked PF (descriptive)	1.335	1.119	1.327	1.410	1.382	1.426	1.309	1.290	1.241
mirror (none within X), research PF	1.135	1.091	1.065	1.032	1.072	1.080	1.095	1.108	1.087

This is a hump, not a gradient. By band, research R/trade is +0.046 (0-1 ATR), +0.010 (1-2), +0.034 (2-3), **-0.055 (3-4)**, **-0.082 (4-6)**, +0.048 (6+). The 3 ATR ceiling is doing real work and it was chosen on research; the locked ladder reproduces the shape (2-4 ATR the plateau) but the specialist should treat 3.0 as one rung of a 2-4 plateau, not a constant.

5. Drop-one: is it the single print or "below yesterday's high"?

A prior-session single print above the close is inside yesterday's range, so the condition co-selects *close below the prior RTH high* 100% of the time. That is the inverse of a gate this branch shipped elsewhere (STUDY_V17), so it had to be separated.

CONDITION	KEEP %	RESEARCH N	PF	CONTROL P	LOCKED N	PF	P (DESCRIPTIVE)
close above prior RTH high (V17 gate)	39	319	1.144	0.345	162	1.034	0.708
close below prior RTH high	61	514	1.202	0.155	252	1.209	0.158
below prior high AND single print ≤ 3 ATR above	37	350	1.446	0.000	187	1.382	0.017
below prior high AND no single print ≤ 3 ATR	24	230	1.035	0.705	94	0.809	0.967
below prior high AND a single print > 3 ATR	20	186	1.021	0.700	81	0.820	0.942

CONDITION	KEEP %	RESEARCH N	PF	CONTROL P	LOCKED N	PF	P (DESCRIPTIVE)
below prior TPO VAH AND single print ≤ 3	23	235	1.512	0.000	116	1.766	0.000
above prior TPO VAH AND single print ≤ 3	14	151	1.263	0.140	89	0.887	0.883

Below yesterday's high without a near single print loses on both blocks. The last two rows are post-hoc splits of the locked block and are recorded, not selected on: the 1.766 cell is exactly the kind of spike a second locked read manufactures.

6. Everything else that was tested

6a. Entry conditions — 43 declared, each vs a same-selectivity random filter, BH q 0.10

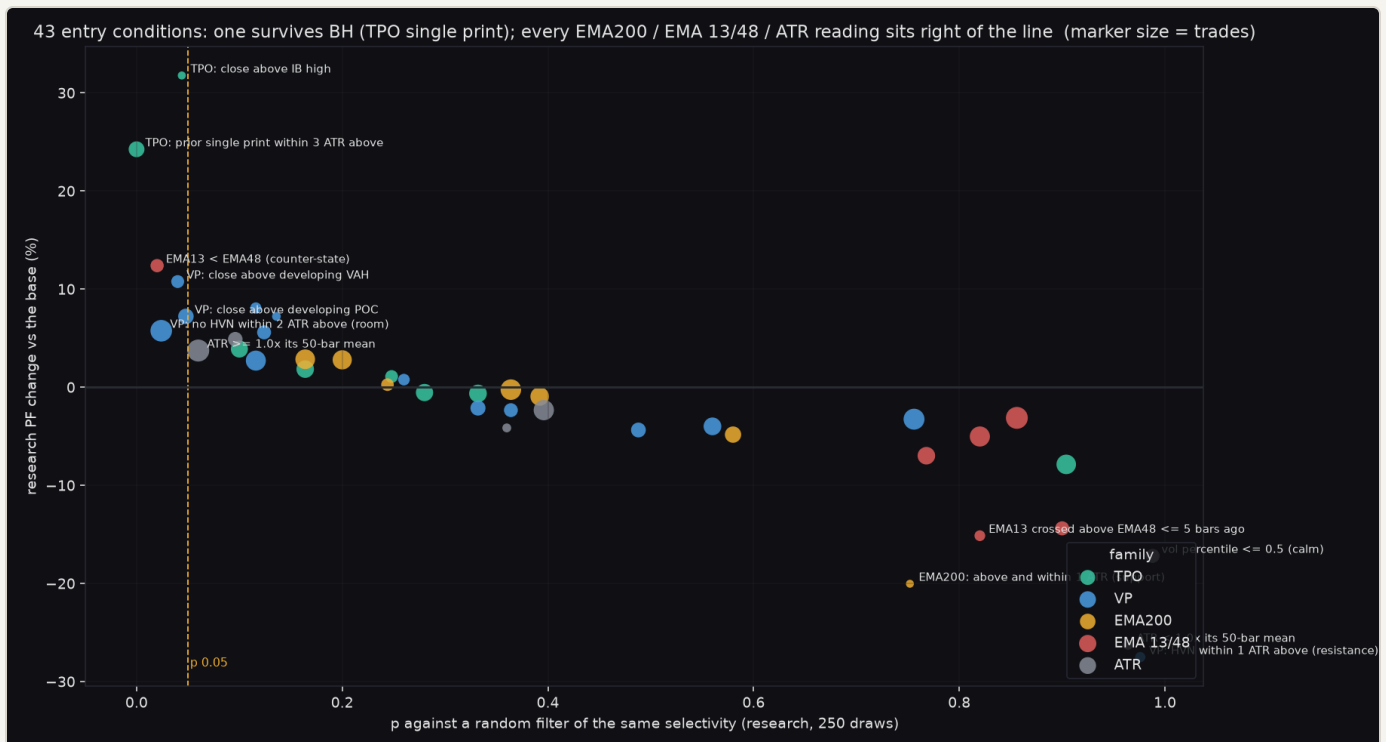


Fig 5. Research PF change vs control p-value; marker size is trade count. 6 of 41 scorable at $p \leq 0.05$ against 2.1 expected; one passes BH.

FAMILY	CELLS	MEAN Δ PF	MEAN P	SHARE BEATING CONTROL
TPO	8	+6.7%	0.26	25%

FAMILY	CELLS	MEAN ΔPF	MEAN P	SHARE BEATING CONTROL
Volume profile	14	+0.3%	0.31	21%
EMA200	7	-2.9%	0.39	0%
EMA 13/48	6	-5.4%	0.70	17%
ATR	6	-6.9%	0.48	0%

- **EMA200 as support/resistance.** Above 1.161 (p 0.36); ≥ 1 ATR above 1.197 (p 0.16); rising 1.196 (p 0.20); touched within 5 bars 1.107 (p 0.58); below 1.167 (p 0.24). The literal support reading — above and within 1 ATR — is the family's worst cell, **PF 0.931**.
- **EMA 13/48 as momentum.** The bullish state passes 74% of breakout bars (the trigger restated) and reads 1.105 (p 0.82); fresh cross ≤ 5 bars 0.988; spread ≥ 0.5 ATR 1.082; spread rising 1.127. The counter-state (13 below 48) is the only reading that clears, 1.308 at p 0.020, and fails BH. It was not read on locked.
- **Volume profile.** Close above the developing VAH 1.289 (p 0.040) and developing POC 1.248 (p 0.048) clear research and fail BH; an HVN within 1 ATR above is the worst VP cell at 0.844; prior-session VAH/POC/VAL do nothing.
- **ATR regime.** Floors +4 to +5% at p 0.06-0.10; ceilings -17 to -26%. The four ATR features that "passed" the IC test are one signal (p 0.98 between them) and it is the ATR-denominator effect, not a forecast.

CONDITION	KEEP %	N	PF	ΔPF %	SHARPE	TOTAL %	CONTROL PF	P
TPO: prior single print within 3 ATR above	37.1	350	1.446	+24.25	2.84	+16.42	1.127	0.000
EMA13 < EMA48 (counter-state)	25.7	247	1.308	+12.39	2.00	+8.72	1.094	0.020
VP: no HVN within 2 ATR above (room)	80.4	653	1.231	+5.75	1.66	+17.44	1.153	0.024
VP: close above developing VAH	20.3	234	1.289	+10.76	1.86	+9.68	1.091	0.040
TPO: close above IB high	6.3	92	1.533	+31.77	2.93	+6.43	1.074	0.044
VP: close above developing POC	32.2	335	1.248	+7.22	1.65	+11.74	1.111	0.048
ATR ≥ 1.0 x its 50-bar mean	81.7	675	1.207	+3.73	1.53	+17.05	1.155	0.060
ATR ≥ 1.2 x its 250-bar mean	29.1	297	1.221	+4.89	1.52	+9.76	1.100	0.096

CONDITION	KEEP %	N	PF	ΔPF %	SHARPE	TOTAL %	CONTROL PF	P
TPO: prior profile skewed low (POC in lower half)	54.0	406	1.209	+3.89	1.55	+10.03	1.131	0.100
VP: developing VA narrow (< 1.5 ATR)	12.8	174	1.258	+8.07	1.58	+5.45	1.074	0.116
VP: close above prior POC	72.3	561	1.195	+2.71	1.48	+12.92	1.141	0.116
VP: naked POC within 3 ATR above	27.1	275	1.229	+5.59	1.52	+7.98	1.106	0.124
VP: close below prior VAL	11.6	107	1.248	+7.22	1.59	+3.03	1.080	0.136
TPO: IB range >= 1.5 ATR	42.3	430	1.185	+1.84	1.30	+10.90	1.126	0.164
EMA200: >= 1 ATR above (floor)	70.2	539	1.197	+2.84	1.50	+12.42	1.140	0.164
EMA200: rising (10-bar slope > 0)	67.5	515	1.196	+2.78	1.51	+11.54	1.144	0.200
EMA200: close below (resistance side)	23.4	223	1.167	+0.25	1.19	+4.59	1.093	0.244
TPO: close above developing VAH	16.9	229	1.176	+1.09	1.17	+6.09	1.086	0.248
VP: prior VA narrow (< 3 ATR)	19.0	189	1.173	+0.77	1.22	+4.77	1.098	0.260
TPO: single print within 2 ATR below (support)	37.5	417	1.157	-0.55	1.13	+9.17	1.114	0.280
VP: close above prior session high	38.9	318	1.139	-2.13	1.04	+5.67	1.111	0.332
TPO: close above prior VAH	54.6	437	1.156	-0.65	1.15	+8.44	1.130	0.332
signal bar range < 0.8 ATR	6.8	111	1.115	-4.15	0.76	+1.46	1.038	0.360
VP: LVN within 1 ATR above	25.7	270	1.136	-2.34	0.96	+4.62	1.096	0.364
EMA200: close above	76.6	589	1.161	-0.25	1.23	+11.24	1.143	0.364
EMA200: >= 2 ATR above (floor)	63.1	482	1.153	-0.95	1.17	+8.70	1.138	0.392
signal bar range >= 1.5 ATR	49.7	576	1.137	-2.34	0.98	+9.84	1.117	0.396

CONDITION	KEEP %	N	PF	ΔPF %	SHARPE	TOTAL %	CONTROL PF	P
VP: close inside prior value area	33.1	307	1.113	-4.37	0.79	+4.06	1.109	0.488
VP: close above prior VAH	55.3	441	1.117	-4.00	0.89	+6.50	1.129	0.560
EMA200: touched within 5 bars	43.7	372	1.107	-4.84	0.84	+5.48	1.124	0.580
EMA200: above and within 1 ATR (support)	6.4	89	0.931	-20.04	-0.51	-0.88	1.071	0.752
VP: prior VA wide (≥ 3 ATR)	81.0	613	1.126	-3.26	0.97	+8.83	1.150	0.756
EMA13-48 spread ≥ 0.5 ATR	56.5	431	1.082	-6.99	0.61	+4.42	1.124	0.768
EMA13 crossed above EMA48 ≤ 5 bars ago	18.1	163	0.988	-15.14	-0.09	-0.29	1.082	0.820
EMA13 $>$ EMA48 (state)	74.3	566	1.105	-5.03	0.80	+7.30	1.146	0.820
EMA13-48 spread rising (5 bars)	87.0	669	1.127	-3.13	1.00	+10.77	1.155	0.856
EMA13 crossed above EMA48 ≤ 20 bars ago	34.1	278	0.996	-14.39	-0.03	-0.14	1.107	0.900
TPO: no prior single print within 3 ATR above	62.9	534	1.072	-7.87	0.53	+4.86	1.133	0.904
ATR $< 1.0\times$ its 50-bar mean	18.3	145	0.858	-26.25	-1.10	-2.02	1.078	0.964
VP: HVN within 1 ATR above (resistance)	11.9	153	0.844	-27.51	-1.21	-3.39	1.072	0.976
vol percentile ≤ 0.5 (calm)	37.0	280	0.964	-17.19	-0.29	-0.99	1.108	0.988

6b. Take profit — 37 rules, fixed ladder vs profile levels, each level vs a random target at the same distance



Fig 6. Left: the fixed ATR target ladder with every profile-level target plotted at its median distance. Right: the fixed stop ladder in three units, with the non-fixed stop rules at their median risk.

Fixed ladder on research: 1.0 ATR 0.944, 1.5 0.997, 2.0 1.091, **2.3 1.164**, 3.0 1.149, 4.0 1.019, 6.0 1.099, none 1.069. Every level rule reads 1.03–1.14. The IB high clears a random target from its own distance distribution (p 0.065, 0.030 capped) — it is a real level — and still loses to the fixed multiple. **0 of 37 beat fixed on PF and total while clearing their null.**

TARGET RULE	N	PF	WIN %	TOTAL %	R/TRADE	SHARPE	DD %	TARGET HIT %	MEDIAN DIST ATR	RANDOTARGET PF
fixed 1.0 ATR	999	0.944	66.9	-4.83	-0.0343	-0.51	7.77	67	1.00	
fixed 1.5 ATR	886	0.997	59.3	-0.26	-0.0172	-0.02	5.86	59	1.50	
fixed 2.0 ATR	802	1.091	53.9	+8.43	+0.0096	0.75	4.19	51	2.00	
fixed 2.3 ATR	765	1.164	51.5	+15.04	+0.0255	1.26	5.52	48	2.30	
fixed 3.0 ATR	706	1.149	44.6	+13.85	+0.0200	1.09	4.85	38	3.00	
fixed 4.0 ATR	669	1.019	37.7	+1.87	-0.0146	0.15	6.87	25	4.00	
fixed 6.0 ATR	618	1.099	34.0	+8.98	+0.0204	0.67	8.54	17	6.00	

TARGET RULE	N	PF	WIN %	TOTAL %	R/TRADE	SHARPE	DD %	TARGET HIT %	MEDIAN DIST ATR	RANDOM TARGET PF
no target (stop / channel / hold only)	565	1.069	29.9	+5.71	+0.0204	0.43	8.01	0		
prior VP VAH (else 2.3 ATR)	746	1.077	48.5	+7.23	+0.0095	0.59	5.93	44	2.30	1.112
prior VP VAH (else no target)	629	1.035	36.4	+3.17	+0.0110	0.24	7.47	17	3.44	1.074
prior VP VAH capped 4 ATR (else 2.3)	783	1.110	52.2	+10.32	+0.0162	0.87	5.33	49	2.30	1.140
nearest prior HVN above (else 2.3 ATR)	760	1.120	50.7	+10.98	+0.0103	0.92	5.86	47	2.30	1.120
nearest prior HVN above (else no target)	615	1.050	36.6	+4.28	+0.0035	0.32	7.09	14	2.33	1.046
nearest prior HVN above capped 4 ATR (else 2.3)	777	1.133	52.1	+12.21	+0.0109	1.03	5.31	50	2.30	1.138
nearest naked POC above (else 2.3 ATR)	693	1.068	44.9	+5.94	-0.0007	0.48	6.87	35	2.30	1.124
nearest naked POC above (else no target)	632	1.028	37.8	+2.40	-0.0073	0.19	6.51	18	4.99	1.063

TARGET RULE	N	PF	WIN %	TOTAL %	R/TRADE	SHARPE	DD %	TARGET HIT %	MEDIAN DIST ATR	RANDOM TARGET PF
nearest naked POC above capped 4 ATR (else 2.3)	780	1.108	51.8	+10.08	+0.0061	0.86	6.05	48	2.30	1.125
prior TPO VAH (else 2.3 ATR)	745	1.115	48.5	+10.61	+0.0138	0.86	5.63	43	2.30	1.114
prior TPO VAH (else no target)	626	1.063	36.3	+5.62	+0.0108	0.42	7.28	16	3.52	1.086
prior TPO VAH capped 4 ATR (else 2.3)	784	1.140	52.4	+12.93	+0.0180	1.09	4.90	49	2.30	1.141
prior TPO POC (else 2.3 ATR)	752	1.108	50.1	+9.86	+0.0119	0.82	6.42	47	2.30	1.126
prior TPO POC (else no target)	604	1.026	33.8	+2.26	+0.0017	0.17	7.19	11	2.70	1.047
prior TPO POC capped 4 ATR (else 2.3)	774	1.123	52.1	+11.29	+0.0147	0.96	5.87	49	2.30	1.143
prior TPO single print above (else 2.3 ATR)	763	1.098	50.2	+9.12	+0.0091	0.76	5.11	45	2.30	1.104
prior TPO single print above (else no target)	645	1.093	39.5	+8.09	+0.0231	0.62	6.20	20	2.75	1.089

TARGET RULE	N	PF	WIN %	TOTAL %	R/TRADE	SHARPE	DD %	TARGET HIT %	MEDIAN DIST ATR	RANDOM TARGET PF
prior TPO single print above capped 4 ATR (else 2.3)	784	1.082	51.9	+7.77	+0.0034	0.66	4.97	49	2.30	1.105
developing TPO VAH (else 2.3 ATR)	789	1.102	52.9	+9.47	+0.0059	0.82	5.62	50	2.30	1.099
developing TPO VAH (else no target)	578	1.075	32.4	+6.13	+0.0170	0.47	7.68	4	1.08	1.069
developing TPO VAH capped 4 ATR (else 2.3)	789	1.102	52.9	+9.47	+0.0059	0.82	5.62	50	2.30	1.101
developing TPO single print above (else 2.3 ATR)	776	1.126	52.4	+11.56	+0.0134	0.99	5.61	49	2.30	1.115
developing TPO single print above (else no target)	569	1.077	31.1	+6.38	+0.0226	0.48	8.01	2	0.94	1.073
developing TPO single print above capped 4 ATR (else 2.3)	776	1.126	52.4	+11.56	+0.0134	0.99	5.61	49	2.30	1.113
IB high (else 2.3 ATR)	806	1.139	54.8	+12.46	+0.0125	1.10	4.16	52	2.30	1.103

TARGET RULE	N	PF	WIN %	TOTAL %	R/TRADE	SHARPE	DD %	TARGET HIT %	MEDIAN DIST ATR	RANDOM TARGET PF
IB high (else no target)	601	1.108	36.4	+8.49	+0.0231	0.66	5.34	12	1.07	1.068
IB high capped 4 ATR (else 2.3)	806	1.141	54.8	+12.69	+0.0132	1.12	4.16	52	2.30	1.098
nearest of {VP VAH, HVN, TPO VAH, single print} (else 2.3)	791	1.087	52.8	+8.13	+0.0032	0.68	5.46	50	2.30	1.108
nearest level if within 2 ATR (else 2.3)	799	1.089	53.6	+8.39	+0.0030	0.71	5.52	51	2.30	1.099

6c. Stop — 27 rules in three units

The fixed ladder peaks at **3.0 ATR** in %/trade, R and return/drawdown together (rare on this branch, where the three units usually disagree). Every structural stop under a profile level is at or below the fixed multiple it falls back to, because at this geometry it mostly is the fallback (median risk 3.19 in every structural row). "Below EMA200" ties 3.0 on research and reads 1.082 on locked. The V22 adaptive direction (wide when calm) beats its inversion on every line and does not beat fixed.

STOP RULE	N	PF	WIN %	%/TRADE	R/TRADE	TOTAL %	DD %	RET/DD	STOPPED %	MEI RIS ATR
fixed 1.00 ATR	903	0.980	29.6	-0.0017	-0.0699	-1.53	6.72	-0.23	71	1.0
fixed 1.50 ATR	825	1.103	40.0	+0.0104	+0.0026	+8.59	4.64	1.85	60	1.5
fixed 2.00 ATR	783	1.134	46.2	+0.0150	+0.0249	+11.71	5.49	2.13	54	2.0

STOP RULE	N	PF	WIN %	%/TRADE	R/TRADE	TOTAL %	DD %	RET/DD	STOPPED %	MEI RIS ATR
fixed 2.50 ATR	774	1.141	49.1	+0.0165	+0.0187	+12.80	4.93	2.60	51	2.5
fixed 3.00 ATR	767	1.173	51.2	+0.0206	+0.0287	+15.79	5.62	2.81	48	3.0
fixed 3.19 ATR	765	1.164	51.5	+0.0197	+0.0255	+15.04	5.52	2.72	48	3.1
fixed 4.00 ATR	764	1.139	51.6	+0.0171	+0.0147	+13.05	6.03	2.16	47	4.0
fixed 6.00 ATR	764	1.116	51.7	+0.0146	+0.0054	+11.14	7.00	1.59	47	6.0
adaptive: 3.19 if vol pct <= 0.5 else 2.0 (V22 direction)	775	1.153	47.6	+0.0173	+0.0344	+13.39	5.77	2.32	52	2.0
adaptive: 4.0 if vol pct <= 0.5 else 2.5 (V22 direction)	770	1.163	49.9	+0.0190	+0.0299	+14.66	4.93	2.97	49	2.5
adaptive: 2.5 if vol pct <= 0.5 else 1.5 (V22 direction)	805	1.136	43.1	+0.0144	+0.0277	+11.56	5.56	2.08	57	1.5
adaptive INVERTED: 2.0 if vol pct <= 0.5 else 3.19	772	1.143	50.1	+0.0171	+0.0159	+13.19	5.56	2.37	50	3.1
adaptive INVERTED: 1.5 if vol pct <= 0.5 else 2.5	793	1.108	45.9	+0.0123	-0.0056	+9.75	4.22	2.31	54	2.5
2.0 ATR x (ATR/ATR50)	778	1.133	49.2	+0.0157	-0.0053	+12.24	5.74	2.13	50	2.6

STOP RULE	N	PF	WIN %	%/TRADE	R/TRADE	TOTAL %	DD %	RET/DD	STOPPED %	MEI RIS ATR
-- wider when ATR is expanding										
3.19 ATR x (ATR/ATR50) -- wider when ATR is expanding	765	1.118	51.5	+0.0147	-0.0028	+11.28	6.65	1.70	47	4.2
2.0 ATR / (ATR/ATR50) -- wider when ATR is contracting	816	1.052	38.6	+0.0052	+0.0117	+4.28	6.76	0.63	61	1.4
3.19 ATR / (ATR/ATR50) -- wider when ATR is contracting	779	1.128	47.0	+0.0146	+0.0363	+11.35	6.19	1.84	53	2.4
1.0 x signal- bar range (floor 1 ATR, cap 6)	822	1.104	42.6	+0.0111	+0.0094	+9.15	6.92	1.32	57	1.6
1.5 x signal- bar range (floor 1 ATR, cap 6)	784	1.106	48.1	+0.0126	+0.0210	+9.87	7.03	1.40	51	2.4
2.0 x signal- bar range (floor 1 ATR, cap 6)	768	1.105	50.5	+0.0130	+0.0209	+9.98	6.31	1.58	48	3.3
below prior VP VAL + 0.25 ATR (fallback 3.19, [1,6])	770	1.145	50.6	+0.0173	+0.0216	+13.32	4.99	2.67	49	3.1
below prior VP POC + 0.25 ATR	770	1.142	50.1	+0.0169	+0.0191	+13.02	6.58	1.98	49	3.1
below developing	770	1.152	49.6	+0.0179	+0.0264	+13.80	6.05	2.28	50	3.1

STOP RULE	N	PF	WIN %	%/TRADE	R/TRADE	TOTAL %	DD %	RET/DD	STOPPED %	MEI RIS ATR
VP VAL + 0.25 ATR										
below TPO IB low + 0.25 ATR	767	1.139	50.8	+0.0169	+0.0148	+12.97	6.02	2.15	48	3.1
below nearest HVN below + 0.25 ATR	775	1.143	49.7	+0.0169	+0.0150	+13.11	6.68	1.96	50	3.1
below EMA200 + 0.25 ATR	767	1.175	51.0	+0.0206	+0.0316	+15.79	5.11	3.09	48	3.1
below EMA48 + 0.25 ATR	780	1.110	48.2	+0.0131	-0.0065	+10.21	6.32	1.62	51	3.1

6d. Feature correlation on the signal bars

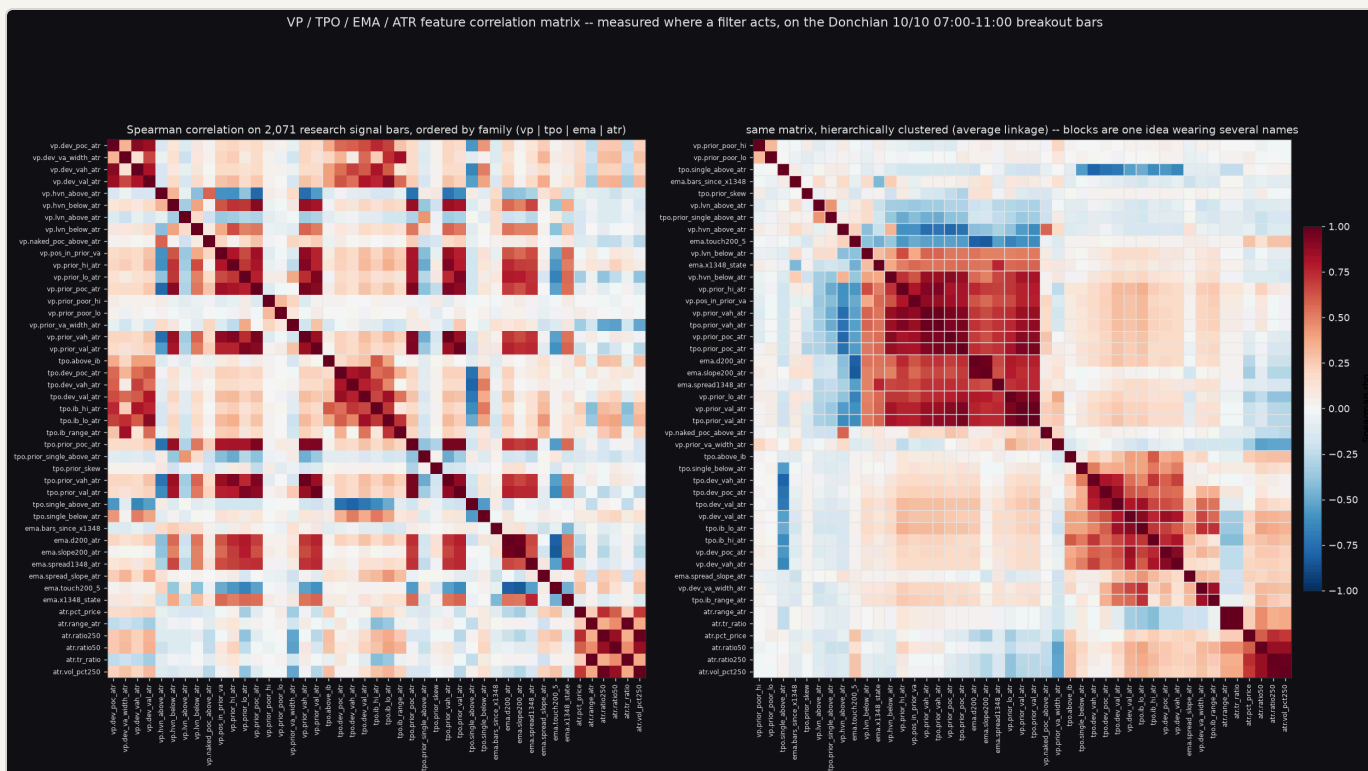


Fig 7. Spearman correlation on the 2,071 research breakout bars, by family and clustered. Twenty pairs at $|\rho| \geq 0.9$: the VP and TPO value areas are the same levels (0.98), the five prior-session distances are one distance, the ATR family is one block (0.98), EMA200 distance and slope are one (0.965). The survivor's strongest partner is the prior high at -0.49 .

7. Implementation and parity

- **Order model.** Signal on a closed 15m bar; fill at the next bar's open plus one tick; stop = fill – 3.19 ATR(signal bar), target = fill + 2.3 ATR, both live from the fill bar; channel exit = 10-bar low, capped at the prior close; stop and target in the same bar resolve as a stop. That ambiguous case is **2.25%** of base trades and 2.98% of gated trades, so the tie-break is not what makes the number. A per-bar-stop / per-bar-target walker (vp_tpo.walk_tp) reproduces the engine exactly on 1,154 trades and is what the level targets and variable stops ran through.
- **Pine.** pine/inst/VP_TPO_SCALP_strategy.pine builds the TPO profile from the chart's own bars (30-minute letters, 2.5-point bins on an absolute grid, frozen at 16:00 New York or at the next RTH open if the 16:00 bar is missing). A bar-by-bar Python transliteration of its arrays agrees with the research feature on **100.00% of 70,685 bars** for the single print, POC, VAH, VAL and IB high, and the gate agrees on 100.000% (31,151 true bars both ways). The bracket is placed with the entry so the fill bar is never naked.
- **Two defects the audits caught before any number was read.** (1) Prior-session values were assigned only on days that already had RTH bars, so a pre-open bar read differently under truncation (8 of 12 probes failed). (2) Early-close sessions (13:00 closes, ten in the sample) were never frozen because their last bar was the day's last bar; the next Monday read Wednesday's profile. Found by the Pine parity diff at 99.4%.
- **Files.** research/inst/vp_tpo.py (features, audit, walker), run_vp_scalp.py (battery), run_vp_scalp2.py (ladder, co-selection, mechanism, years, bootstrap), run_vp_scalp3.py (drop-one), vp_tpo_parity.py, outputs in results/inst/vp_scalp*.txt, trade log with the gate flag per trade in results/inst/vp_scalp_trades.csv, study in docs/ib/STUDY_VP_TPO_SCALP.md.

8. Caveats that stay attached

- **One survivor of 43** at BH q 0.10 (6 at $p \leq 0.05$ against 2.1 expected), on **one market** and **one geometry**. The branch has seen five perfect research plateaus fail out of sample; one locked read is one replication.
- **Locked n = 187.** Day-block bootstrap $P(\text{mean} \leq 0) = \mathbf{0.056}$, 95% CI $[-0.012, +0.122]$ %/trade. It clears its matched control (p 0.016) and does

not quite clear zero. Those are two different questions and both are reported.

- **Drawdown.** Realised locked 4.7% of price; permutation places it at the 45th percentile of its own reshuffles; MC p95 10.5%, **p99 13.1% (2.8× realised)**. At one MNQ near 20,000 that is roughly \$5,200. Size for the p99.
- **Sample regime.** NQ rose 89% over the file; the strategy is long-only; the short mirror was not tested.
- **Execution.** Spread is assumed (no bid/ask in the feed). The base survives 2× the assumed spread on this branch's earlier read; the gated version has not been re-stressed.
- **Untested robustness axes.** Bin size (2.5 points) and letter length (30 minutes) were fixed by convention, never swept. The 15-minute chart is the only timeframe tested.

9. Decisions for the specialist

These are the calls that would move the result, in the order I would take them. None was run, so each is a genuinely open question.

1. **Cross-market read of the TPO half.** US100 and US30 exist here as 15-minute feeds (nine and eight years, 2016–2025, including 2018, COVID and the 2022 bear) and the TPO profile needs only 15-minute bars. Freeze the rule exactly as shipped, run it on both, and read their locked blocks once. A level feature that works on the Nasdaq only is a regime; one that holds on the Dow is a mechanism. This is the single most informative thing left.
2. **Robustness of the profile construction.** Sweep bin size (1, 2.5, 5, 10 points and an ATR-scaled bin) and letter length (15, 30, 60 minutes) on research only. If the single-print condition needs 2.5-point bins to work, it is an artefact of resolution; if it survives 5 and 10, it is a structural feature.
3. **Widen the sample rather than the pool.** The base rate of the condition is 37% and its lift is 0.91, so it is not starved. The bootstrap at $n = 187$ is the binding constraint; another 12 months of forward data (post 2025-12-12) would settle $P(\text{mean} \leq 0)$ faster than any further filter search. Recommend paper or minimum-size forward trading with the trade log format in §7 and a pre-registered stopping rule (e.g. abandon if the forward PF over 100 trades falls below 1.0).

4. **Interaction with value area.** The post-hoc split "below prior VAH and single print above" reads 1.51 research / 1.77 locked on n 116, and "above VAH and single print above" reads 0.89 locked. If the specialist wants to pursue it, pre-register it and read it on US100/US30, never again on NQ's locked block, which is now spent for this family.
5. **Short side.** The mirrored condition (single print within 3 ATR *below* after a downside breakout) is a two-line change on the feature module. Expect it to fail on this sample's drift; it is still the cheapest falsification available.
6. **Do not spend effort on:** the EMA200 support reading, any bullish EMA 13/48 reading, profile levels as targets, structural stops, or a tighter stop. All four axes were swept and the marginals point the other way.

Every number in this document is reproducible from the scripts named in §7 on commit acdc974 or later of the branch. Study text: docs/ib/STUDY_VP_TPO_SCALP.md. Prior context on the same base: docs/ib/STUDY_BAYESOPT_SCALP.md, docs/ib/STUDY_SCALP_FILTERS.md. Repository conventions and the full list of prior findings: CLAUDE.md.